Server-to-Edge Ranking Transfer for Telecom LLMs

AI-generated under human Research Director prompting for the CNSM 2026 "Agentic AI Researcher" experimental track

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Abstract—Telecom operators use language models to interpret intents, answer standards questions, and support control decisions, but an edge deployment must also meet limits on latency, memory, and energy. Because most telecom studies rank candidates on data-center GPUs, we test whether those rankings lead to the same choice on edge hardware. Our pre-registered study evaluates 23 model configurations on an NVIDIA Jetson AGX Orin using fixed samples from 6G-Bench and TeleQnA, scoring each model's actual output and measuring energy for every request. Accuracy rankings transfer well from an A100 reference (ρ=0.92), making server results useful for shortlisting, but they do not identify the best available edge deployment. A mixture-of-experts model absent from the source study's published ranking beats its server winner by 4.0 points (p=0.022) while using about half the energy. The published Edge Score transfers less strongly (ρ=0.67), and we also find that Q4 is usually slower than Q8, one model collapses to chance across four precisions, and the most energy-efficient power mode depends on how idle power is counted. Server benchmarks can therefore provide a shortlist, but the final artifact and runtime must be tested on the target device. Code and supporting artifacts are available in a public GitHub repository.

Index Terms—edge computing, large language models, 5G/6G network management, energy measurement, benchmarking, replication

I. Introduction

Language models are becoming part of automated 5G/6G operations, where they translate operator intents, retrieve standards knowledge, and support control decisions. Choosing a model for the edge therefore requires more than an accuracy comparison because the model must also respond quickly, fit in memory, and stay within an energy budget.

Most evidence for this choice comes from server hardware: recent telecom studies use eight A100s [1], an L40S [2], or an RTX A6000 [3], yet none reports energy per query, and the A100 study warns that its results should not be interpreted as edge performance.

An edge device is not simply a smaller server: its CPU and GPU may share memory, its power modes trade speed against power draw, and its quantized kernels may behave differently. On Jetson hardware, for example, INT8 can be slower than FP16 [4], so these platform differences may change which model should be deployed.

We test that possibility on a Jetson AGX Orin by reproducing the roster and ranking method of the A100 Edge-Score study [1], adding Q4 and Q8 configurations from the L40S study [2], and running two telecom benchmarks. For every deployed artifact, we score the output it actually produces and measure its energy use on the device.

The main result is that server accuracy ranks can identify promising models but cannot certify an edge deployment. Across 49 runs, we find that the model artifact and runtime can change accuracy, quantization can save memory without saving time, and the best power mode depends on whether idle power is counted; the study supports these findings with pre-registered transfer tests and a complete table of measured joules per query.

II. Prior Evidence and Study Questions

The main server reference evaluates ten instruction-tuned models on 6G-Bench and ranks them with an Edge Score, $ES = A/(L \cdot M)$, that combines accuracy, latency, and VRAM on A100 GPUs [1]. A second study tests Q4 and Q8 GGUF models through Ollama/llama.cpp on an L40S [2], while other telecom work reports accuracy without energy [3], compares local models with cloud API prices [5], or assumes that local inference has no cost [6].

Studies on physical edge devices measure power modes, energy, or deployment costs [4], [7]–[11], but they use general-purpose tasks. Telecom studies that come closest use server-class hardware, cloud-backed agents, or very small test sets without energy measurements [12]–[14]. In our survey of 138 works, we found no study that measures both telecom-task accuracy and energy per query on a power-constrained edge device, a gap also identified in recent work [2], [15].

We ask three questions: RQ1 examines whether server rankings, including the top choice, transfer to the Jetson; RQ2 tests whether expected relationships among precision, latency, and memory hold in a unified-memory runtime; and RQ3 asks how workload, model architecture, and power mode affect energy per query.

Before collecting Jetson results, we froze four expectations: C-E1 predicts strong transfer of accuracy ranks ($\rho \geq 0.7$) but weak transfer of Edge Score ($\rho < 0.5$), while C-E2 predicts at least one latency or energy inversion caused by quantization. C-E3 predicts sublinear energy growth and the best accuracy per joule at 1.5–3 B parameters, and C-E4 requires a complete joules table.

III. Experimental Design

A. Device and Model Configurations

We use a Jetson AGX Orin Developer Kit with 32 GB of unified memory, L4T R36.4.4, and CUDA 12.6. Main runs use the 30 W power mode, while a pre-registered ablation repeats one Qwen2.5-3B configuration at 15 W and MAXN. We verify the active mode before every run and observe peak temperatures of 54 °C in 30 W runs and 64 °C at MAXN, both below throttling. Inference uses llama.cpp commit 22dc605c, built for CUDA sm_87, with full GPU offload, a 4096-token context, one slot, and no prompt-cache reuse. Model files are identified by SHA-256 hashes; for multipart files, the hash covers the first shard, while the runtime is pinned by commit rather than by a binary hash.

The planned roster contains 24 configurations from 135 M to 8.3 B parameters, covering the A100 models at their intended deployment precision, matched F16 and Q8 runs, and additional Q4/Q8 pairs. It also includes LFM2-8B-A1B, a mixture-of-experts (MoE) model with 8.3 B total parameters but about 1.5 B active per token; because this model is absent from the source paper's published ranking, we label comparisons that include it. Olmo-3-7B is excluded because its chat template fails in the pinned runtime, and a post-hoc Llama-3.2-1B BF16 diagnostic is excluded from pre-registered statistics.

B. Workloads and Output Scoring

We take fixed-seed, task-stratified samples of 1,200 questions from 6G-Bench [16] and 1,200 from TeleQnA [17], [18], analyzing the two suites separately. The first three questions of each run are warm-ups, leaving the same 1,197 scored questions for every model; recomputing the A100 reference on this subset does not change its ranking.

For 6G-Bench, we reproduce the official prompt and parser. For TeleQnA, we accept only a JSON answer or an explicit "answer": N. This strict rule corrects version 1, in which echoed option numbers gave SmolLM2-135M a small false score; no other result changes to four decimal places. Decoding is greedy with a 16-token limit and each model's GGUF chat template. We do not constrain output with a grammar because format failures are part of the server baseline.

C. Energy, Memory, and Transfer Tests

We sample three INA3221 rails at 20 Hz—GPU/SOC, CPU/CV, and the 5 V system input—and define each request's gross energy as the trapezoidal integral of their summed power. Net energy subtracts the same run's idle baseline of 4.6–5.1 W; gross energy matters when a battery powers the whole board, whereas net energy isolates the added inference cost on an always-on device. We report both as rail measurements rather than wall- or battery-terminal measurements.

Latency is reported as per-request p50 and p95, while peak memory is the server process's VmHWM, including weights, cache, and buffers in shared memory. For the A100 comparison, we recompute accuracy from released per-question records and use the published latency and VRAM [1]; we analyze the L40S comparison separately [2]. Pre-registered tests include rank correlations (Spearman ρ and Kendall τ), top-choice agreement, Wilson confidence intervals, and permutation tests. We retain raw outputs, parsed answers, timings, power traces, temperatures, model hashes, and runtime details, and deterministic scripts generate every reported result. The reproducibility package is available at https://github.com/dl423/cnsm2026-server-to-edge-telecom-llms.

IV. Results

Table I summarizes the main configurations, with additional precision variants supporting the comparisons below.

A. Server Ranks Shortlist Models but Do Not Preserve the Choice

Across the ten shared models, Jetson accuracy largely follows the A100 ordering (Fig. 1(a)), with a strong rank correlation ($\rho = 0.92$, $p < 0.001$) despite being lower by 5.8 points on average. Server accuracy is therefore useful for building a shortlist.

However, it does not identify the best model once we expand the roster. Qwen2.5-7B, the A100 winner, scores 0.641 on the Jetson, whereas LFM2-8B-A1B scores 0.681—a significant 4.0-point gain over the same 1,197 questions (exact McNemar $p = 0.022$)—while answering twice as fast and using about half as much energy under either gross or net accounting.

This result requires one important qualification: LFM2-8B-A1B is not in the source paper's published ranking, and we reconstructed its server accuracy from released records. Within the original published roster, Qwen2.5-7B remains the accuracy winner on both systems, while the published Edge-Score winner, LFM2-350M, also retains its position. Device testing thus changed the best available deployment but not the winner within the original roster.

The original Edge Score transfers only moderately ($\rho = 0.67$), so C-E1 is only partly supported. This comparison is descriptive because the two systems use different measures: the server score uses published mean latency and peak VRAM, whereas the Jetson score uses p50 latency and process VmHWM. Transfer from the L40S study is stronger for both accuracy ($\rho = 0.81$) and

TABLE I
Main on-device results at 30 W. We report pass@1, median latency, gross (J_g) and net (J_n) energy per query, and peak process memory. Wilson 95% CI half-widths are $\leq \pm 0.028$ (n=1197). Chance is 0.25 for 6G-Bench and 0.219 for TeleQnA. †Strict parser; v1's lenient parser credited echoed digits (0.038).

Model	B	Quant	6G-Bench (n=1197 scored) pass@1	t_{p50}	J_g	J_n	TeleQnA (n=1197 scored) pass@1	t_{p50}	J_g	J_n	GB
SmolLM2-135M	0.14	F16	0.226	0.47	5.1	2.9	0.000†	0.30	3.1	1.7	8.4
LFM2-350M	0.35	Q4_K_M	0.337	0.68	7.4	4.2	0.427	0.19	1.9	1.0	8.4
Granite-4.0-h-350M	0.35	Q4_K_M	0.227	1.17	13.5	7.9	0.277	0.30	2.9	1.5	8.7
Granite-4.0-h-1B	1.00	Q4_K_M	0.532	3.23	37.9	22.4	0.542	0.77	8.5	4.8	8.7
LFM2.5-1.2B	1.20	Q4_K_M	0.469	1.78	20.4	11.6	0.549	0.37	4.4	2.6	8.5
Llama-3.2-1B	1.24	F16	0.226	1.28	18.5	11.8	0.282	0.39	5.9	4.0	8.9
Qwen2.5-1.5B	1.54	Q4_K_M	0.512	2.22	25.2	14.5	0.496	0.47	5.3	3.0	8.4
Qwen2.5-3B	3.09	Q4_K_M	0.543	4.17	48.0	27.2	0.505	0.82	9.9	6.0	8.4
Llama-3.2-3B	3.21	Q4_K_M	0.611	4.12	47.1	25.6	0.582	0.85	10.2	6.1	8.6
Gemma-3-4B	4.30	Q4_K_M	0.593	5.67	63.5	34.9	0.602	1.64	19.3	11.8	9.0
Gemma-3-4B	4.30	Q8_0	0.565	5.13	60.8	34.8	0.607	1.63	22.5	14.8	9.2
Qwen2.5-7B	7.62	Q4_K_M	0.641	8.41	97.2	57.6	0.668	1.51	19.8	12.5	8.6
LFM2-8B-A1B (MoE)	8.30	Q4_K_M	0.681	4.12	49.0	29.9	0.614	0.88	10.7	6.5	8.6

TABLE II
Ranking transfer. ‡LFM2-8B-A1B is absent from the source paper's published ranking; we derive its server accuracy from released per-question records.

Ranking comparison	n	ρ	p	τ	top agr.
Accuracy (A100 → Jetson)	10	0.92	0.000	0.81	no
w/o LFM2-8B-A1B‡	9	0.91	0.001	0.82	yes
Edge Score $A/(L{\cdot}M)$	10	0.67	0.039	0.56	yes
Server ES → EES $A/(L{\cdot}E)$	10	0.50	0.146	0.38	no
Accuracy, F16 matched	4	0.63	0.500	0.55	yes
Accuracy (L40S overlap)	10	0.81	0.007	0.64	no
Warm latency (L40S)	10	0.90	0.001	0.72	no

latency ($\rho = 0.90$), perhaps because both systems use the llama.cpp lineage.

Replacing memory with measured energy gives an Energy-Edge Score, $EES = A/(L \cdot E)$, under which several models move by at least three ranks (Fig. 1(b)). Because this score ranks the failed SmolLM2-135M first merely because it is cheap to run, an efficiency score should be applied only after setting a minimum accuracy requirement.

B. The Inference Stack Can Dominate Precision

Two small models show why literal outputs matter. SmolLM2-135M mostly answers "A," stays near chance on 6G-Bench, and produces no valid TeleQnA answers; at F16, it chooses "A" on 99.4% of 6G-Bench questions. Because its server result is equally poor, this behavior indicates a model failure rather than a Jetson-only failure.

Llama-3.2-1B behaves differently: its accuracy falls from 0.376 on the A100 to 0.226 on the Jetson, where it answers "A" on 99.9–100% of questions at BF16, F16, Q8, and Q4. We verified a byte-matched prompt template and the tokenization, making a prompt or conversion error unlikely. Although the failure is associated with the artifact–runtime combination, all four precisions use the same runtime, so we cannot isolate the cause; this is a reproducible association rather than a proven mechanism. Testing several precisions in one runtime is therefore not enough to certify a model.

C. Q4 Saves Capacity but Usually Does Not Save Time

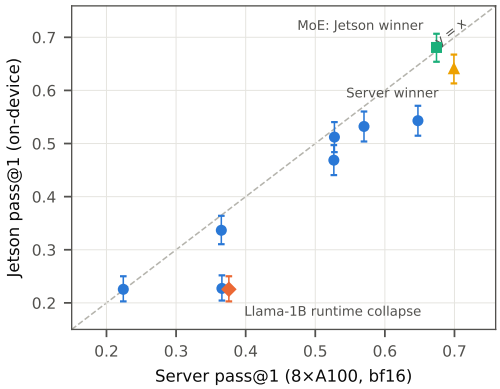
The matched runs contain 21 within-model precision inversions and one cross-model energy-order flip, supporting C-E2. Q4 is slower than Q8 for every complete model family on 6G-Bench: for Qwen2.5-7B, median latency rises from 7.40 s at Q8 to 8.41 s at Q4, while at 1 B, Q4 is also slower than F16 (1.56 versus 1.28 s). Long-prompt prefill and dequantization overhead are a likely explanation.

Q4 nevertheless saves memory and sometimes energy: for the 7.6 B model, it reduces memory from 15.1 to 8.6 GB and net energy from 82 to 58 J without changing accuracy or latency. Its accuracy effects also depend on the task, as Granite-1B Q8 trails Q4 by 4.4 points on 6G-Bench but leads it by 3.1 points on TeleQnA. On this device, Q4 is therefore a memory optimization rather than a reliable speed optimization.

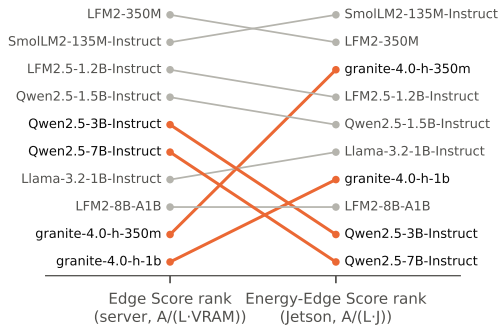
D. Energy Depends on Workload, Architecture, and Power Mode

For Q4 models, net energy ranges from 2.8 to 57.6 J per 6G-Bench query and gross energy from 5.3 to 97.2 J, while the 7B F16 run reaches 82.2 J net. TeleQnA uses only 0.9–12.5 J net because its prompts are shorter, showing that joules per query are meaningful only when the workload and accounting method are stated.

A 56-fold parameter range produces a 20-fold net-energy range (Fig. 2(a)), so energy grows more slowly than parameter count, as C-E3 predicted, but the expected 1.5–3 B efficiency optimum is not supported. A raw accuracy-per-joule ratio instead favors the failed SmolLM2-135M because it uses so little energy; version 1 removed such models with a post-data filter and therefore reported the wrong optimum. When we instead require accuracy within five points of the best result, LFM2-8B-A1B is preferred at 0.681 accuracy and 29.9 J net, demonstrating that efficiency must be evaluated alongside a minimum capability requirement.

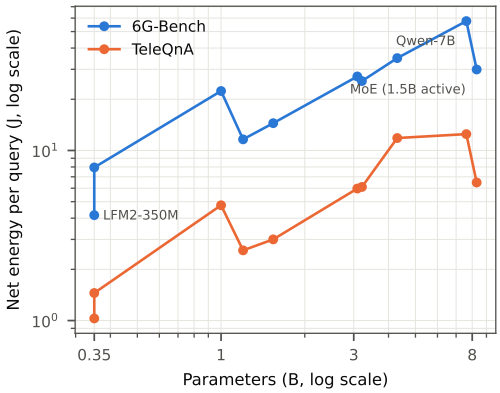


(a) Accuracy transfer

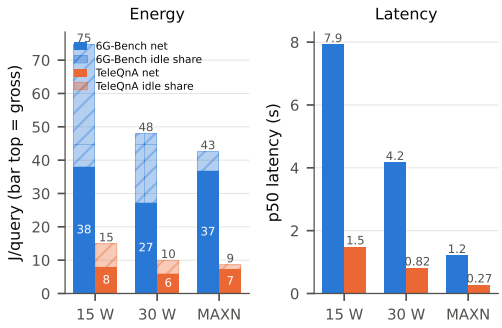


(b) Efficiency-rank transfer

Fig. 1. Server-to-device transfer. (a) 6G-Bench accuracy for the ten shared models. Whiskers show Wilson 95% intervals; the dashed line marks equal accuracy. (b) Server Edge Score versus Jetson Energy-Edge Score. Orange lines mark rank changes of at least three places.



(a) Energy scaling across Q4 models



(b) Power-mode ablation

Fig. 2. Energy on the Jetson. (a) Net energy per query at 30 W. LFM2-8B-A1B has 8.3 B total parameters but only 1.5 B active per token. (b) Qwen2.5-3B at three power modes. Solid bars show net energy; the full bar shows gross energy. The best mode changes with the accounting method.

Power mode provides a second example of why accounting matters (Fig. 2(b)). Net energy is 27.2 J at 30 W, 38.0 J at 15 W, and 36.8 J at MAXN, whereas gross energy gives a different order: MAXN is lowest at 42.6 J, followed by 30 W at 48.0 J and 15 W at 74.7 J. MAXN finishes in 1.21 s, versus 4.17 s at 15 W, so its higher power draw is repaid by shorter execution, with accuracy changing by at most three questions. An always-on board should therefore use 30 W to minimize added inference energy, whereas a busy board, or one that sleeps between requests, should use MAXN to finish sooner and minimize total energy.

E. Unified Memory and the Cloud Anchor

On 6G-Bench, process memory stays near 8.3–9.2 GB for models up to 4 B, and the server retains about 47 MiB from each long prompt until its default cache fills, even though reuse is disabled. This cache competes with model weights on a unified-memory device: some models use only 0.9–2.5 GB on the shorter TeleQnA prompts, while others remain near 9 GB, and for 7B F16, about 14 GB of weights dominates both workloads. Discrete-GPU VRAM is therefore a poor proxy for unified-memory demand.

Claude Haiku 4.5 provides a cloud reference, scoring 0.756 on the TeleQnA subset; the best edge configuration, Qwen2.5-7B Q4, reaches 0.668, or 88% of that accuracy, at 12.5 J net (19.8 J gross) per query and no API fee. Haiku is not scoreable on 6G-Bench because its prose preamble consumes the 16-token budget before producing valid JSON, yielding a score of 0.001; even a 512-token diagnostic leaves more than 99% of answers invalid. The API calls cost $4.44.

V. Deployment Procedure and Limitations

The results suggest a two-stage deployment process: first use server benchmarks to shortlist accurate models, and then test the exact model file, runtime, prompt, and power mode on the intended device. This second stage must measure energy directly and inspect actual outputs rather than relying only on aggregate scores.

The device test should begin with a capability gate, because a candidate that stays near chance, repeatedly

emits one option, or fails the output parser should not be rescued by a favorable efficiency ratio. Models that pass this gate can then be compared on latency, memory, and energy using both long and short prompts, since prefill changes all three measures. Certification should also record the exact artifact hash, runtime revision, chat template, parser failures, and power mode, and it should compare power modes under both gross and net energy. This order prevents a cheap but unusable model from becoming the apparent deployment winner.

Energy estimates must also state what they include: Granite-1B, for example, uses about 22 J net and 38 J gross per 6G-Bench query, so a 10 Wh battery supports about 1,600 queries if idle power is budgeted separately but only about 950 if the battery powers the whole board. At one query per minute, the roughly 5 W idle load reduces this estimate to about 115 queries.

Limitations. We test one board, one llama.cpp version, and two benchmarks, and TeleQnA may appear in model training data. Energy comes from three internal rails rather than a wall or battery meter, while each configuration is run once, preventing us from measuring variation across runs, boards, or thermal conditions.

The server and Jetson comparisons also differ in precision and in their latency and memory measures. The A100 reference uses BF16, whereas most deployment runs use Q4; four matched F16/BF16 configurations only partly separate hardware and precision effects. Olmo-3-7B is excluded, and ranking tests cover only ten models, so threshold conclusions remain sensitive to the roster. Finally, the raw data store question identifiers and outputs rather than complete prompts. Reproduction therefore depends on the pinned samples, chat templates, model hashes, runtime commit, and deterministic scripts.

VI. Conclusion

Server benchmarks can identify promising telecom LLMs, but they cannot certify an edge deployment: on the Jetson, accuracy ranks transfer well even though an expanded roster produces a different best model. Q4 saves memory but is often slower than Q8, and a model can fail across several precisions when all use the same runtime. Moreover, the best power mode changes depending on whether idle power is counted. The practical lesson is therefore to test the final artifact on the intended hardware, set a minimum accuracy requirement, and measure both net and gross energy on the real workload.

Appendix

Disclosure Statement

This paper was generated entirely by AI systems, which performed the literature review, research ideation, experimental design and execution, data analysis, and manuscript preparation. Claude Fable 5, accessed through Claude Code, was the primary AI researcher; Claude Opus 5 and Claude Sonnet 5 were used to a lesser extent. GPT-5.6 Sol, accessed through Codex, reviewed and revised an earlier draft for clarity and accessibility.

The initial human prompt specified the broad topic of "5G/6G Networks and Edge Computing Services" and asked the AI to survey current literature and identify gaps. Claude Code proposed several research directions and, after the human Research Director selected one, conducted the experiments, analyzed the results, and drafted and revised the manuscript. It was informed that its host machine provided access to a Jetson Orin GPU. Codex critiques were returned to Claude Code for self-correction, and Codex later improved the draft's readability. The human Research Director monitored progress and provided intermediate direction through prompting.

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